Computer Assignment: Chow-Liu Tree Learning

Overview. In this assignment you will implement the Chow-Liu pipeline for learning a tree-structured distribution on binary variables from data. The goal is to connect ideas from graphical models, mutual information, maximum spanning trees, parameter estimation, and statistical evaluation. You will work with the provided notebook `exercise.ipynb`, which contains scaffolding for synthetic-data generation, tree utilities, experiment-running code, and plotting. Your job is to complete the core estimation functions and then study how performance changes with sample size.

Setup and notation. Let $X = (X_1, \ldots, X_n)$ with each $X_i \in \{0, 1\}$. Let $T = (V, E)$ be a tree on vertex set $V = \{0, 1, \ldots, n-1\}$. After choosing a root r, orient every edge away from the root, so the distribution takes the form

$$p(x) = p_r(x_r) \prod_{(u,v) \in E^*} p_{v|u}(x_v \mid x_u),$$

where E^* is the set of directed edges after orientation. Thus a tree-structured model is specified by one root marginal and one 2×2 conditional table for each directed edge. Suppose we observe ℓ i.i.d. samples

$$x^{(1)}, x^{(2)}, \ldots, x^{(\ell)} \in \{0, 1\}^n.$$

The Chow-Liu estimator first computes empirical pairwise mutual informations, then chooses a maximum-weight spanning tree, and finally fits the tree parameters from empirical counts.

Required implementation functions. Please implement exactly the following functions in the notebook:

1. `empirical_pairwise_mi(samples, eps=1e-12)`
2. `estimate_tree_distribution(samples, edges, root=0, alpha=1.0)`
3. `kl_true_to_estimate(true_params, est_params, eps=1e-15)`

All other code in the notebook is provided as scaffolding or reference code.

1. **Tree factorization and parameterization.** Explain why a rooted tree-structured binary distribution can be written as

   $$p(x) = p_r(x_r) \prod_{(u,v) \in E^*} p_{v|u}(x_v \mid x_u).$$

   How many free scalar parameters are needed to specify such a model on a tree with n nodes? Compare this with the number of free parameters needed for a completely general distribution on $\{0, 1\}^n$. Briefly explain why the tree model is computationally attractive.

2. **Pairwise mutual information and the Chow-Liu objective.** For two binary random variables X_i and X_j, write the pairwise mutual information

$$I(X_i; X_j) = \sum_{a=0}^{1} \sum_{b=0}^{1} p_{ij}(a,b) \log \frac{p_{ij}(a,b)}{p_i(a)p_j(b)}.$$

   Then define the empirical mutual information $\widehat{I}_{ij}$ using the observed samples. Explain the Chow-Liu idea: estimate all pairwise mutual informations, then recover a maximum spanning tree whose edge weights are the values $\widehat{I}_{ij}$. You do not need to provide a full proof, but you should explain clearly why large mutual information suggests that an edge is useful in the learned tree.

3. **Empirical MI implementation.** Implement `empirical_pairwise_mi`. Use the notebook signature `(samples, eps=1e-12)`. The function should use the equation from step 2 to compute all pairwise MI terms, which will be organized in the off-diagonal of an $n \times n$ matrix. Use empirical marginals and empirical joint frequencies, and include the supplied `eps` parameter only for numerical stability in log-ratio calculations.

   What information theoretic quantity would this formula produce on the diagonal? Will it be needed fo the Chow-liu algorithm? If not, what constant value may be used to replace the estimates obtained for the diagonal?

   In your writeup, state the computational complexity of your implementation as a function of the number of variables n and the number of samples ℓ.

4. **Maximum spanning tree recovery.** The notebook includes a Kruskal implementation for a maximum spanning tree. Explain how Kruskal's algorithm works in this setting and why it returns exactly $n-1$ edges. Also explain why a spanning tree, rather than an arbitrary dense graph, is the correct model class for the Chow-Liu estimator.

5. **Parameter estimation on a fixed tree.** Implement `estimate_tree_distribution`. Use the notebook signature `(samples, edges, root=0, alpha=1.0)`. This function should:

   (a) orient the recovered tree away from the chosen root,

   (b) estimate the root marginal probability $P(X_r = 1)$ from counts,

   (c) estimate each conditional table $P(X_v = x_v \mid X_u = x_u)$ from counts,

   (d) apply Laplace smoothing with constant `alpha`,

   (e) return the parameter dictionary in the same format used by the supplied sampling code.

   State the formulas you use for the smoothed root estimate and the smoothed conditional estimates.

6. **KL divergence by exact enumeration.** Implement `kl_true_to_estimate(true_params, est_params, eps=1e-15)`. This function should compute

$$\mathrm{KL}(P_{\mathrm{true}} \| P_{\mathrm{est}}) = \sum_{x \in \{0,1\}^n} p_{\mathrm{true}}(x) \log \frac{p_{\mathrm{true}}(x)}{p_{\mathrm{est}}(x)}.$$

   For the dimensions in the notebook, this can be done by enumerating all 2^n binary assignments. Explain briefly why this exact computation is reasonable for the provided experiment sizes, but would become infeasible for much larger n. This quantity can actually be computed much more efficiently using local marginals. As an optional challenge, implement the faster method and use the slow method to verify your results.

7. **One-trial visualization and diagnosis.** Use the visualization cell to compare one true tree and one recovered Chow-Liu tree, together with the true and estimated mutual-information matrices. Discuss what failure looks like when the tree is not recovered correctly. If you see recovery errors at small sample sizes, explain whether the estimated MI matrix appears noisy, whether the incorrect edges have similar weights to the true ones, and how this affects the KL divergence.

8. **Smoothing study.** Repeat part of the experiment for at least two different smoothing values, for example $\alpha = 0$ and $\alpha = 0.5$ or $\alpha = 1.0$. Describe how smoothing changes the estimated parameters and whether it appears to help or hurt KL divergence for small sample sizes. Be specific and support your comments with plots or tables.

9. **Deliverables.** Please submit:

   (a) the completed notebook,

   (b) a short writeup answering the conceptual questions above,

   (c) the key plots from your experiments,

   (d) a brief summary of what you learned about tree recovery, mutual-information estimation, and parameter estimation from finite samples.

   Your writeup should be concise but complete. It should explain the formulas you implemented, summarize the experimental behavior, and comment on any interesting failure modes or implementation choices.